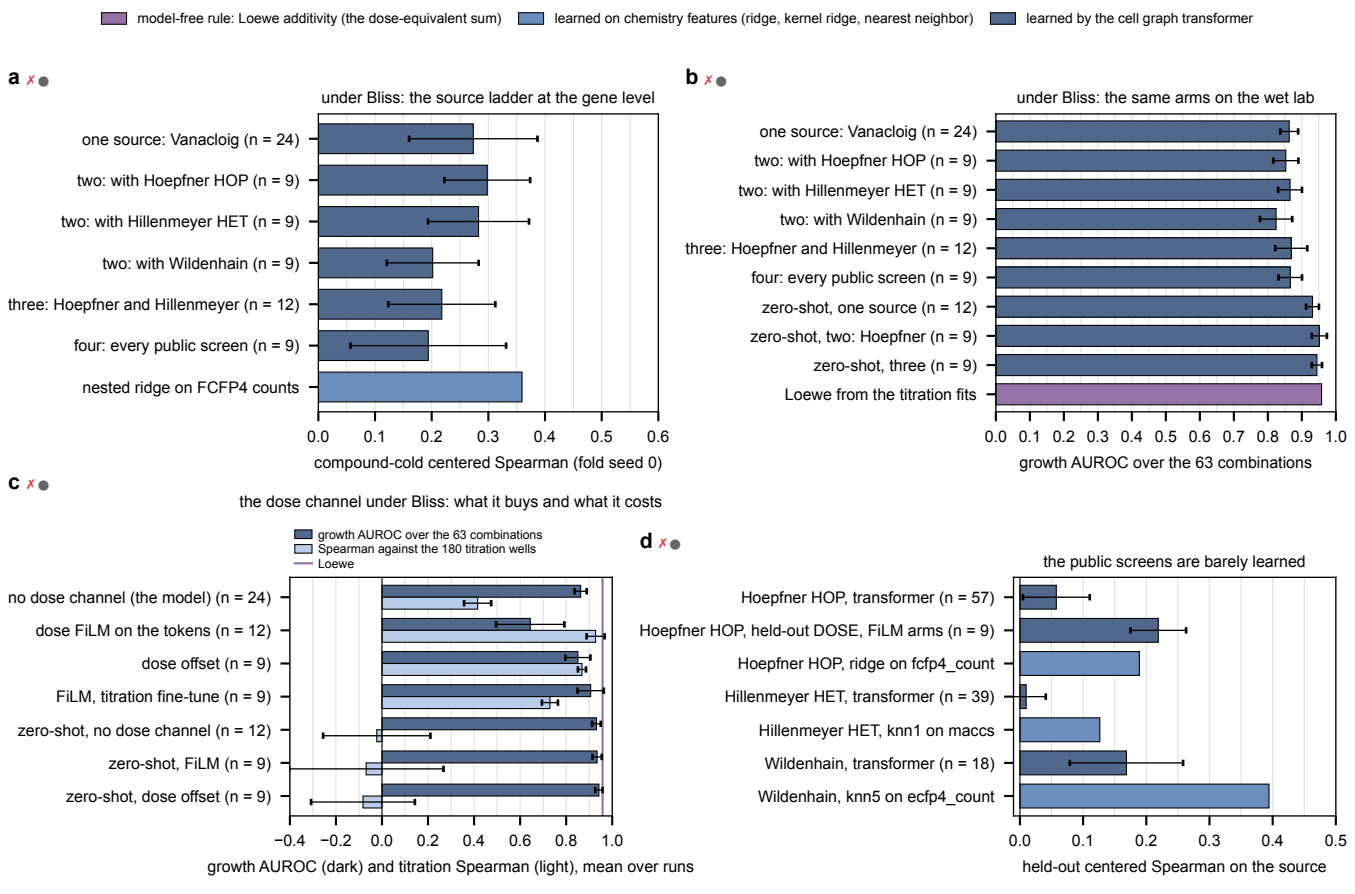
model-free rule: Loewe additivity (the dose-equivalent sum)
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
under Bliss: the source ladder at the gene level
one source: Vanacloig (n = 24)
two: with Hoepfner HOP (n = 9)
two: with Hillenmeyer HET (n = 9)
two: with Wildenhain (n = 9)
three: Hoepfner and Hillenmeyer (n = 12)
four: every public screen (n = 9)
nested ridge on FCFP4 counts
0.0 0.1 0.2 0.3 0.4 0.5 0.6
compound-cold centered Spearman (fold seed 0)
b
under Bliss: the same arms on the wet lab
one source: Vanacloig (n = 24)
two: with Hoepfner HOP (n = 9)
two: with Hillenmeyer HET (n = 9)
two: with Wildenhain (n = 9)
three: Hoepfner and Hillenmeyer (n = 12)
four: every public screen (n = 9)
zero-shot, one source (n = 12)
zero-shot, two: Hoepfner (n = 9)
zero-shot, three (n = 9)
Loewe from the titration fits
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
growth AUROC over the 63 combinations
c
the dose channel under Bliss: what it buys and what it costs
growth AUROC over the 63 combinations
Spearman against the 180 titration wells
Loewe
no dose channel (the model) (n = 24)
dose FiLM on the tokens (n = 12)
dose offset (n = 9)
FiLM, titration fine-tune (n = 9)
zero-shot, no dose channel (n = 12)
zero-shot, FiLM (n = 9)
zero-shot, dose offset (n = 9)
−0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0
growth AUROC (dark) and titration Spearman (light), mean over runs
d
the public screens are barely learned
Hoepfner HOP, transformer (n = 57)
Hoepfner HOP, held-out DOSE, FiLM arms (n = 9)
Hoepfner HOP, ridge on fcfp4_count
Hillenmeyer HET, transformer (n = 39)
Hillenmeyer HET, knn1 on maccs
Wildenhain, transformer (n = 18)
Wildenhain, knn5 on ecfp4_count
0.0 0.1 0.2 0.3 0.4 0.5
held-out centered Spearman on the source